

ELSA: Acoustic Event-Wise Semantic Alignment for Fine-Grained Reference-Free Text-to-Audio Evaluation

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Abstract

Text-to-audio (TTA) generation, which synthesizes audio directly from natural language descriptions, has attracted growing attention due to its capability reflecting detailed user intent across speech, sound effects, and music. For the development of TTA models, reliable evaluation of generated audio is essential. While human subjective evaluation remains the gold standard, it is costly and time-consuming, motivating the use of automatic evaluation metrics. In this paper, we propose ELSA, a reference-free evaluation metric for TTA generation that enables fine-grained text–audio comparison. ELSA decomposes an input text query into semantically distinct acoustic events and performs event-wise alignment with corresponding audio segments, allowing for precise assessment of transient events. Experiments on four TTA benchmarks demonstrate that ELSA consistently achieves higher correlation with human subjective ratings compared to prior metrics.

Index Terms: text-to-audio, automatic evaluation metric, fine-grained semantic similarity, CLAPScore

1. Introduction

Audio generation conditioned on user intent, encompassing speech, sound effects, and music, has been widely studied [1, 2] due to its broad range of applications, including the construction of audio environments for augmented and virtual reality [3] and the synthesis of background music and sound effects for media content [4]. Among these approaches, text-to-audio (TTA) generation, which synthesizes audio directly from arbitrary text descriptions, has attracted considerable attention due to its capability of precisely reflecting detailed user intent. [5, 6].

For the effective development and benchmarking of TTA models, reliable evaluation of generated audio is essential. Human subjective evaluation of both overall audio quality (OVL) and relevance to the input text (REL) is widely regarded as the gold standard [5]. However, despite its high reliability, human evaluation is inherently costly and time-consuming, which has motivated research into automatic evaluation metrics that correlate with human subjective ratings [7, 8].

Automatic evaluation metrics for TTA can be broadly categorized into reference-based metrics [9, 8] and reference-free metrics [7, 10], depending on whether paired reference audio is required. Reference-based metrics enable fine-grained evaluation by directly comparing audio signals. For example, AudioBERTScore [8] computes frame-level similarity in the feature space of an audio foundation model, thereby enabling implicit assessment at the level of individual acoustic

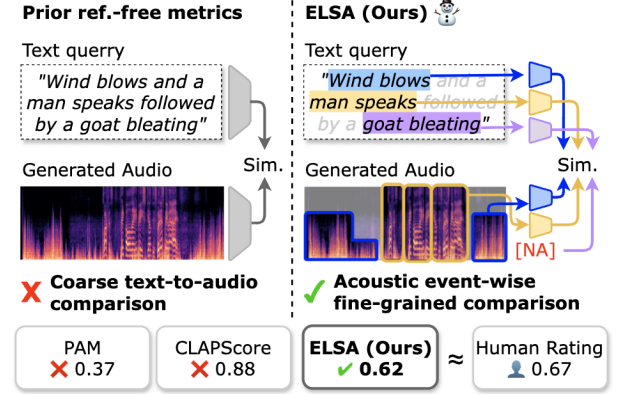


Figure 1: Overview of ELSA. ELSA enables acoustic event-wise, fine-grained text–audio alignment for reference-free TTA evaluation, yielding high correlation with human subjective ratings.

events. However, the requirement for reference audio substantially limits their applicability in real-world scenarios. In contrast, reference-free metrics compare the input text with the generated audio directly, making them applicable to a wider range of experimental settings, including real-world scenarios. Nevertheless, because these methods compare fundamentally different modalities, text and audio, their evaluations are typically restricted to coarse-grained similarity measured in a jointly aligned text–audio embedding space, as exemplified by CLAP-based metrics [10]. As a result, their correlation with human subjective ratings remains limited. For instance, on the RELATE dataset, the Spearman correlation between CLAPScore and human subjective ratings of REL is only 0.280 [11].

To this end, we propose ELSA², a reference-free evaluation metric for TTA generation that enables fine-grained text–audio comparison. Figure 1 presents the core idea of the proposed approach. Unlike existing methods that rely on global text–audio matching, ELSA decomposes the text query into semantically distinct acoustic events and performs event-wise alignment with corresponding audio segments. This design allows for a more precise assessment of transient acoustic events (e.g., footsteps, door slams, or gunshots) that are often overlooked by coarse-grained metrics [10]. Experimental results show that ELSA consistently achieves higher correlation with human subjective ratings than existing metrics across four TTA benchmarks. In addition, ablation and sensitivity analyses demonstrate the contribution of individual components and the robustness of ELSA with respect to the number of target acoustic events.

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¹Project page: <https://ttaevalprojectpage-w83rz.kinsta.page/>

²Acoustic Event-Level Semantic Alignment score.

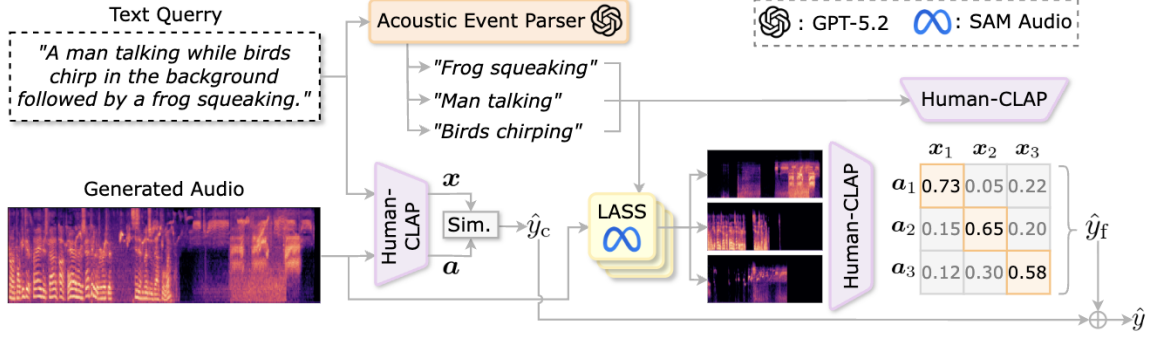


Figure 2: Architecture of ELSA. ELSA embeds the input text query and generated audio into a shared representation space, extracts event-level text and audio representations using a text parser and a language-queried source separation model, and hierarchically aggregates global and event-level similarities for fine-grained, reference-free TTA evaluation.

2. Method

We propose ELSA, a reference-free automatic evaluation metric for TTA generation that enables fine-grained comparison between text and audio. An overview of the proposed metric is shown in Fig. 2. ELSA is inspired by prior metrics developed for video and image captioning, which hierarchically evaluate visual–language alignment [12, 13].

2.1. Event-Wise Representation Extraction

In ELSA, fine-grained text–audio comparison is achieved by extracting multi-granular representations from a text query and the generated audio from an arbitrary TTA model. Specifically, given a text query $\mathbf{x} \in \mathbb{R}^{V \times N}$ and the corresponding generated audio $\mathbf{s} \in \mathbb{R}^L$, both modalities are first embedded into a shared text–audio feature space using a pretrained embedder. This yields global text and audio representations, $\mathbf{t} \in \mathbb{R}^d$ and $\mathbf{a} \in \mathbb{R}^d$, respectively. Throughout our metric, we consistently employ Human-CLAP [11]³ as the embedder, as it provides a feature space highly correlating with human subjective ratings. Here, V , N , L , and d denote the vocabulary size, maximum token length, audio sequence length, and embedding dimension, respectively. These global representations capture coarse user intent expressed in the text query and the generated audio. However, because both modalities are uniformly embedded irrespective of the temporal duration of individual acoustic events, the contribution of transient events (e.g., footsteps, door slams) is relatively attenuated and may be obscured.

To overcome this limitation, we further extract event-level representations that are independent of event duration. Concretely, a text-only LLM decomposes the input text query $\mathbf{x}$ into a set of semantically distinct acoustic events $\{\mathbf{x}_f^{(i)}\}_{i=1}^M$. Each event description $\mathbf{x}_f^{(i)}$ is then embedded to obtain its corresponding text representation $\mathbf{t}_f^{(i)}$. In parallel, conditioning on $\mathbf{x}_f^{(i)}$, a Language-queried Audio Source Separation (LASS) model is used to extract the corresponding audio segment and its audio representation $\mathbf{a}_f^{(i)}$. Here, we use GPT-5.2⁴ as the LLM and SAM-Audio [14]⁵ as the LASS model. To ensure compatibility with the training distribution of the LASS model, $\mathbf{x}_f^{(i)}$ is restricted to a concise noun–verb phrase (e.g., “dog barking”).

2.2. Hierarchical Alignment Scoring

We jointly evaluate global matching between the text query and the generated audio, as well as fine-grained alignment that reflects event-level user intent to yielding a unified final score. We first compute a coarse text–audio matching score $\hat{y}_c$ over the entire input as

$$\hat{y}_c = \frac{\mathbf{t}^\top \mathbf{a}}{\|\mathbf{t}\| \|\mathbf{a}\|}. \quad (1)$$

To further assess fine-grained correspondence at the acoustic-event level, we evaluate the pairwise cosine similarity $f_{ij} = (\mathbf{t}_f^{(i)})^\top \mathbf{a}_f^{(j)} / (\|\mathbf{t}_f^{(i)}\| \|\mathbf{a}_f^{(j)}\|)$ for all combinations of $\mathbf{t}_f^{(i)}$ and $\mathbf{a}_f^{(j)}$. Based on these similarities, we define event-level precision P_f and recall R_f as

$$P_f = \frac{1}{N} \sum_i \left(f_{ii} - \frac{1}{N} \sum_j f_{ji} \right), \quad (2)$$

$$R_f = \frac{1}{N} \sum_i \left(f_{ii} - \frac{1}{N} \sum_j f_{ij} \right). \quad (3)$$

We then compute the F1-score of P_f and R_f , which serves as the fine-grained matching score $\hat{y}_f$:

$$\hat{y}_f = 2 \frac{P_f R_f}{P_f + R_f}. \quad (4)$$

Finally, we adaptively integrate the coarse and fine-grained scores according to the number of acoustic events, yielding the final evaluation score $\hat{y}$:

$$\hat{y} = \lambda^N \hat{y}_c + (1 - \lambda^N) \hat{y}_f. \quad (5)$$

Here, λ is a normalization factor that balances coarse- and fine-grained evaluation, which we set $\lambda = 0.4$ empirically.

3. Experiments

3.1. Datasets and Data Pre-processing

To evaluate the correlation between the proposed metric and human subjective ratings for TTA generation, we employ four public benchmarks. Specifically, we employ AudioCaps [15],

³<https://github.com/sarulab-speech/Human-CLAP>

⁴<https://openai.com/index/introducing-gpt-5-2/>

⁵<https://github.com/facebookresearch/sam-audio>

Table 1: Correlation with human subjective ratings across baseline metrics. ρ and τ denote Spearman’s rho coefficient and Kendall’s tau coefficient, respectively. **Bold** and underlined values indicate the best and second-best results.

Metrics [%]	AudioCaps				RELATE		MusicCaps				Clotho			
	OVL		REL		REL		OVL		REL		OVL		REL	
	ρ	τ	ρ	τ	ρ	τ	ρ	τ	ρ	τ	ρ	τ	ρ	τ
<i>Reference-based Metrics</i>														
SI-SDR	-2.8	-2.4	-2.9	-2.4	-2.4	-2.0	3.7	3.0	4.8	4.0	-4.4	-3.7	0.2	0.2
FD _{OpenI3}	-6.9	-4.5	-3.1	-1.9	-2.1	-1.5	11.7	8.0	5.9	3.8	-5.1	-3.6	4.7	3.2
KL _{PaSST}	7.5	5.1	11.9	8.0	18.3	12.7	18.8	12.9	3.8	2.8	-7.0	-5.1	16.7	11.5
AudioBERTScore	19.9	14.0	<u>28.7</u>	<u>19.6</u>	17.8	12.3	9.4	6.3	28.1	<u>19.4</u>	20.9	14.2	22.5	15.4
<i>Reference-free Metrics</i>														
PAM	<u>22.9</u>	<u>15.7</u>	17.6	11.7	9.2	6.2	<u>19.7</u>	<u>13.3</u>	16.0	10.8	18.0	12.1	0.6	0.6
CLAPScore _{MS}	6.9	4.5	15.8	10.6	15.2	10.4	8.7	6.0	25.9	17.7	7.6	5.1	27.7	18.8
CLAPScore _{LAION}	19.3	13.6	22.1	15.3	18.3	12.6	2.6	1.8	<u>34.1</u>	<u>23.7</u>	11.7	7.8	24.7	17.0
CLAPScore _{Human}	13.6	9.5	26.7	18.7	<u>31.5</u>	<u>21.7</u>	17.8	12.1	9.1	6.2	<u>21.1</u>	<u>14.7</u>	<u>32.9</u>	<u>22.7</u>
ELSA (Ours)	33.9 (+11.0)	23.5 (+7.8)	46.5 (+17.8)	32.7 (+13.1)	37.9 (+6.4)	26.2 (+4.5)	32.1 (+12.4)	22.4 (+9.1)	36.8 (+2.7)	25.2 (+1.5)	41.2 (+19.9)	28.7 (+14.0)	39.8 (+6.9)	27.5 (+4.8)

Clotho [16], and RELATE [17], and MusicCaps [18] benchmarks. In each benchmark, audio samples are generated from text queries using multiple TTA foundation models (e.g., AudioLDM [19] and Tango [?]) and are subsequently annotated with human subjective ratings. As human evaluation criteria, overall perceptual quality (OVL) and relation of audio to text caption (REL) is provided across all benchmarks to assess semantic consistency between the generated audio and the input text. For AudioCaps, we use the human evaluation test sets collected by Deshmukh et al. [7], while for Clotho, we use the test set collected by Kishi et al [8].

Furthermore, to evaluate the correlation with finer-grained human subjective ratings, we employ inclusion of sound event score (IS) and order of sound event score (OS) in RELATE [17] benchmark and for additional criteria.

For data preprocessing, all audio samples are resampled to 16 kHz and either cropped or zero-padded to a fixed duration of 10 seconds. Subsequently, human subjective ratings are averaged across subjects and are normalized to the range [0, 1] to ensure comparability across benchmarks.

3.2. Baseline Metrics

We compare our proposed metric with eight standard automatic evaluation metrics commonly used for TTA generation tasks. These metrics are broadly categorized into reference-based metrics, which require paired reference audio samples, and reference-free metrics, which do not rely on reference signals and therefore serve as direct baselines for our proposed metric. For reference-based metrics, following Evans et al. [20], we adopt scale-invariant signal-to-distortion ratio (SI-SDR) [9], Kullback–Leibler divergence in the PaSST [21] feature space (KL_{PaSST}), Fréchet distance in the OpenI3 [22] feature space (FD_{OpenI3}), and AudioBERTScore [8]. As reference-free metrics, we employ PAM [7] and CLAPScore [10]. For CLAPScore, we evaluate three variants corresponding to different underlying CLAP models: CLAPScore_{MS}, CLAPScore_{LAION}, and CLAPScore_{Human}, based on MS-CLAP [23], LAION-CLAP [24], and Human-CLAP [25], respectively.

4. Results and Analysis

Table 1 reports the correlation between automatic evaluation metrics and human subjective ratings, measured using Spearman’s rank correlation coefficient (ρ) and Kendall’s rank cor-

Table 2: Ablation study of the LASS model and embedding feature spaces on the Clotho benchmark.

Metrics [%]	LASS Model	Feature Space	OVL		REL	
			ρ	τ	ρ	τ
(i)	SAM Audio	Human-CLAP	41.2	28.7	39.8	27.5
(ii)	AudioSep	Human-CLAP	40.9	28.3	43.9	30.6
(iii)	SoloAudio	Human-CLAP	37.1	25.9	36.5	25.2
(iv)	SAM Audio	MS-CLAP	28.2	19.2	26.4	17.9
(v)	SAM Audio	LAION-CLAP	31.5	21.5	30.4	20.6

relation coefficient (τ), where higher values indicate stronger agreement with human judgments. Overall, ELSA consistently achieved the highest correlation across all benchmarks and evaluation criteria over both reference-based and reference-free baselines, demonstrating the effectiveness of event-level, fine-grained text–audio comparison. In contrast, CLAP-based metrics exhibit notable performance variations across benchmarks, suggesting that coarse global similarity is insufficient to consistently capture detailed text–audio correspondence.

Table 2 presents an ablation study on the Clotho benchmark, examining the effects of the language-queried audio source separation (LASS) model and the embedding feature space. Row (i) corresponds to the full ELSA configuration, which combines SAM-Audio [20] with the Human-CLAP [25] feature space and achieves the highest overall correlation with human subjective ratings. Replacing the LASS model with alternative source separation approaches results in only marginal performance differences, indicating that the choice of LASS model has a limited impact on evaluation performance. In contrast, changing the embedding feature space leads to a substantial degradation in correlation across both OVL and REL, highlighting that the feature space plays a critical role in accurately capturing fine-grained text–audio alignment.

Figure 3 illustrates the sensitivity of metric–human correlation to the number of acoustic events contained in the text query. Across all datasets, the proposed method consistently achieves the highest Kendall’s τ regardless of the number of acoustic events, demonstrating strong robustness to variations in query complexity. In contrast, CLAP-based metrics exhibit substantial performance fluctuations as the number of acoustic events changes, and in some cases show a notable degradation when multiple events are present. This behavior indicates that coarse global matching is sensitive to the compositional complexity

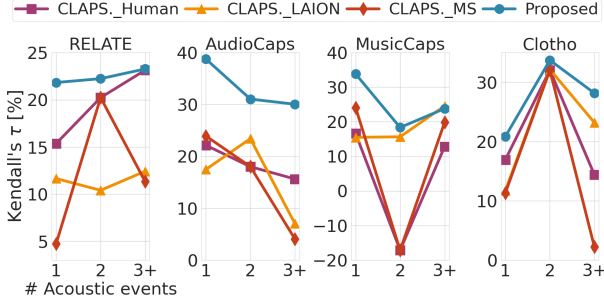


Figure 3: Sensitivity analysis of metric-human correlation with respect to the number of acoustic events in the text query.

Table 3: Performance comparison on fine-grained benchmarks focusing on acoustic event attributes and temporal order.

Metrics [%]	RELATE		CompA			
	IS	OS	Attribute		Order	
	τ	τ	text $\uparrow$	audio $\uparrow$	text $\uparrow$	audio $\uparrow$
PAM	[WIP]	[WIP]	[WIP]	[WIP]	[WIP]	[WIP]
CLAPScore _{MS}	[WIP]	[WIP]	[WIP]	[WIP]	[WIP]	[WIP]
CLAPScore _{LAION}	[WIP]	[WIP]	[WIP]	[WIP]	[WIP]	[WIP]
CLAPScore _{Human}	[WIP]	[WIP]	[WIP]	[WIP]	[WIP]	[WIP]
ELSA (Ours)	00.0 ()	00.0 ()	00.0 ()	00.0 ()	00.0 ()	00.0 ()

of the text query. By explicitly modeling and aligning individual acoustic events, the proposed method maintains stable and superior performance even as the number of target events increases.

These results confirm that event-level alignment enables robust reference-free evaluation of TTA systems across a wide range of text query complexities.

5. Conclusion

In this paper, we proposed ELSA, a reference-free evaluation metric for text-to-audio generation that enables fine-grained text-audio comparison. Experimental results show that ELSA outperforms existing evaluation metrics, including both reference-based and reference-free approaches. Furthermore, sensitivity analyses demonstrate that ELSA is robust to variations in the number of target acoustic events.

[TODO]

- Adding justification for Precision, Recall, and Lambda
- Conducting IS/OS and CompA experiments.
- Polishing every paragraph except for Intro.
- Preparing project page.

6. Acknowledgments

7. Generative AI Use Disclosure

Generative AIs were used exclusively for auxiliary purposes, including language polishing, editing, and manuscript formatting. They were not involved in the research methodology, nor did they contribute to the implementation or evaluation of the proposed methods. Accordingly, Generative AI did not constitute an essential, original, or non-standard component of this research.

8. References

- [1] Liu, Xubo and Iqbal, Turab and Zhao, Jinzheng and Huang, Qiushi and Plumbley, Mark D. and Wang, Wenwu, “Conditional Sound Generation Using Neural Discrete Time-Frequency Representation Learning,” in *MLSP*, 2021, pp. 1–6.
- [2] S. Pascual, G. Bhattacharya, C. Yeh, J. Pons, and J. Serrà, “Full-band General Audio Synthesis with Score-based Diffusion,” in *ICASSP*, 2023, pp. 1–5.
- [3] D. B. Lloyd, N. Raghuvanshi, and N. K. Govindaraju, “Sound synthesis for impact sounds in video games,” in *IBD*, 2011, pp. 55–62.
- [4] S. Forsgren and H. Martiros, “Riffusion: Stable Diffusion for Real-Time Music Generation,” URL: <https://riffusion.com/about>, 2022.
- [5] F. Kreuk, G. Synnaeve, A. Polyak, U. Singer, A. Défossez, J. Copet, D. Parikh, Y. Taigman, and Y. Adi, “AudioGen: Textually Guided Audio Generation,” in *ICLR*, 2023.
- [6] Liu, Haohe and Yuan, Yi and Liu, Xubo and Mei, Xinhao and Kong, Qiuqiang and Tian, Qiao and Wang, Yuping and Wang, Wenwu and Wang, Yuxuan and Plumbley, Mark D., “AudioLDM 2: Learning Holistic Audio Generation With Self-Supervised Pre-training,” *TASLP*, vol. 32, pp. 2871–2883, 2024.
- [7] S. Deshmukh, D. Alharthi, B. Elizalde, H. Gamper, M. Al Ismail, R. Singh, B. Raj, and H. Wang, “PAM: Prompting Audio-Language Models for Audio Quality Assessment,” in *Interspeech*, 2024, pp. 3320–3324.
- [8] M. Kishi, R. Sakai, S. Takamichi, Y. Kanamori, and Y. Okamoto, “AudioBERTScore: Objective Evaluation of Environmental Sound Synthesis Based on Similarity of Audio Embedding Sequences,” in *AAAI Audio-Centric AI Workshop*, 2026.
- [9] J. L. Roux, S. Wisdom, H. Erdogan, and J. R. Hershey, “SDR – Half-baked or Well Done?” in *ICASSP*, 2019, pp. 626–630.
- [10] F. Xiao, J. Guan, Q. Zhu, X. Liu, W. Wang, S. Qi, K. Zhang, J. Sun, and W. Wang, “A Reference-free Metric for Language-Queried Audio Source Separation using Contrastive Language-Audio Pretraining,” in *DCASE Workshop*, 2024.
- [11] T. Takano, Y. Okamoto, Y. Kanamori, Y. Saito, R. Nagase, and H. Saruwatari, “Human-CLAP: Human-perception-based Contrastive Language-audio Pretraining,” *APSIPA ASC*, pp. 131–136, 2025.
- [12] Z. Zeng, J. Sun, H. Zhang, T. Wen, Y. Su, Y. Xie, Z. Wang, and B. Chen, “HICEScore: A Hierarchical Metric for Image Captioning Evaluation,” in *ACM-MM*, 2024, p. 866–875.
- [13] Y. Shi, X. Yang, H. Xu, C. Yuan, B. Li, W. Hu, and Z. Zha, “EMScore: Evaluating Video Captioning via Coarse-Grained and Fine-Grained Embedding Matching,” in *CVPR22*, 2022.
- [14] B. Shi, A. Tjandra, J. Hoffman, H. Wang, Y.-C. Wu, L. Gao, J. Richter, M. Le, A. Vyas, S. Chen, C. Feichtenhofer, P. Dollár, W.-N. Hsu, and A. Lee, “SAM Audio: Segment Anything in Audio,” 2025. [Online]. Available: <https://arxiv.org/abs/2512.18099>
- [15] C. D. Kim, B. Kim, H. Lee, and G. Kim, “AudioCaps: Generating Captions for Audios in The Wild,” in *NAACL-HLT*, 2019.

- [16] K. Drossos, S. Lipping, and T. Virtanen, “Clotho: an Audio Captioning Dataset,” in *ICASSP*, 2020, pp. 736–740.
- [17] Y. Kanamori, Y. Okamoto, T. Takano, S. Takamichi, Y. Saito, and H. Saruwatari, “RELATE: Subjective evaluation dataset for automatic evaluation of relevance between text and audio,” in *Interspeech*, 2025, pp. 3155–3159.
- [18] A. Agostinelli, T. I. Denk, Z. Borsos, J. Engel, M. Verzett, A. Caillon, Q. Huang, A. Jansen, A. Roberts, M. Tagliasacchi *et al.*, “MusicLM: Generating Music From Text,” *arXiv:2301.11325*, 2023.
- [19] H. Liu, Z. Chen, Y. Yuan, X. Mei, X. Liu, D. Mandic, W. Wang, and M. D. Plumbley, “AudioLDM: Text-to-audio generation with latent diffusion models,” in *ICML*, vol. 202, 23–29 Jul 2023, pp. 21 450–21 474.
- [20] Z. Evans, C. Carr, J. Taylor, S. H. Hawley, and J. Pons, “Fast Timing-Conditioned Latent Audio Diffusion,” in *ICML*, 2024.
- [21] Khaled Koutini and Jan Schlüter and Hamid Eghbal-zadeh and Gerhard Widmer, “Efficient Training of Audio Transformers with Patchout,” in *Interspeech*, 2022, pp. 2753–2757.
- [22] A. L. Cramer, H.-H. Wu, J. Salamon, and J. P. Bello, “Look, Listen, and Learn More: Design Choices for Deep Audio Embeddings,” in *ICASSP 2019*, 2019, pp. 3852–3856.
- [23] B. Elizalde, S. Deshmukh, M. A. Ismail, and H. Wang, “CLAP Learning Audio Concepts from Natural Language Supervision,” in *ICASSP*, 2023, pp. 1–5.
- [24] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, “Large-Scale Contrastive Language-Audio Pretraining with Feature Fusion and Keyword-to-Caption Augmentation,” in *ICASSP*, 2023, pp. 1–5.
- [25] T. Takano, Y. Okamoto, Y. Kanamori, Y. Saito, R. Nagase, and H. Saruwatari, “Human-CLAP: Human-perception-based Contrastive Language-audio Pretraining,” in *APSIPA ASC*, 2025, pp. 131–136.